

# Varikuntla Sai Manoj

✉ varikuntlasaimanoj@gmail.com    ☎ 9100161401    🔗 varikuntlasaimanoj    in VarikuntlaSaiManoj

🌐 VARIKUNTLASAIMANOJ

## Professional Summary

Recent CSE(AI/ML) Graduate with 6 months of hands-on internship experience developing Generative AI, Retrieval-Augmented Generation (RAG), LLM-based applications, and deep learning forecasting systems. Experienced in Python, LangChain, LangGraph, Qdrant, Llama models, FastAPI, React, and Streamlit. Built AI systems for scientific knowledge retrieval, agro-meteorological forecasting, sentiment analysis, and multi-agent applications.

## EDUCATION

**B.Tech Computer Science and Engineering - AIML**

Malla Reddy University

Sept 2022 – June 2026

CGPA: 8.09/10

## SKILLS

**Programming & Data:** Python, SQL, Pandas, NumPy, Matplotlib

**AI / Machine Learning:** Machine Learning, Deep Learning, NLP, Generative AI, RAG, LLMs, Sentiment Analysis

**AI Frameworks & LLM Technologies:** LangChain, LangGraph, Llama, Google Gemini, API Integration

**Backend & Frontend:** FastAPI, React.js, Streamlit, HTML, CSS

**Databases & Cloud:** MySQL, Supabase, Vector Databases, AWS

**Tools & Platforms:** Git, GitHub, Postman, Power BI

## Experience

**ICRISAT**

AI/ML Intern

Feb 2026 – Aug 2026

Hyderabad, India

- Built a RAG system using LangGraph, Qdrant, and local Llama models for scientific research retrieval, achieving 80%+ response accuracy.
- Designed a custom 6-layer evidence validation pipeline to reduce hallucinations and improve the reliability of LLM-generated responses without dependency on external APIs.
- Developed an agro-meteorological intelligence platform combining 30-year ERA5 validation, GRU/LSTM forecasting, and FAO-56 irrigation advisory for Telangana.
- Integrated NVIDIA Earth-2, FourCastNet 3, and CorrDiff into a real-time forecasting platform for mandal-level predictions across 33 Telangana districts.

## PROJECTS

**ShopScribe - AI-Powered Shopping Assistants Using Sentiment Analysis**

[ShopScribe](#) 🔗

- Built an AI-powered Streamlit application using Firecrawl, Google Gemini, and VADER to analyze product listings from Amazon, Flipkart, and Myntra.
- Implemented real-time sentiment analysis, trend visualization, and aspect-based insights for e-commerce intelligence.

**Tejoraj**

[Tejoraj](#) 🔗

- Built a full-stack AI space platform using React, FastAPI, and LangGraph, integrating NASA APIs, arXiv, and multi-agent source-cited responses.
- Developed research paper generation, live NASA imagery, PDF/DOC export, and 3D solar-system visualization with configurable reasoning modes.

## PUBLICATIONS

**AI-Powered Shopping Assistants Using Sentiment Analysis**

Apr 2025

Prof. Sabyasachi Chakraborty, V. Sai Manoj, Tejaswi Mahadev, U. Pranay Reddy, T. Vishal Prem, T. Vamshi Krishna

[10.56025/IJARESM.2025.1304252562](#) 🔗

## CERTIFICATIONS & ACHIEVEMENTS

- Udemy Certification: The Complete Python Pro Bootcamp; Complete Generative AI Course With LangChain and HuggingFace
- Postman Certification: API Fundamentals Student Expert
- 2nd Prize — National Level Hackathon, Malla Reddy University
- Winner — AgentX 24-Hour Innovation Hackathon, Anurag University / Salesforce; designed and implemented an AI-driven customer onboarding and CRM workflow using Agentforce and Data Cloud.